BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 078 · Practical Programming Project

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. For valid ages 13–18, give an ordinary normal test value.

Q2. Where can a local variable be used?
A. In every program
B. Within its local scope
C. Across every network
D. In all other subprograms

Q3. “Data interception is only copying a paper document from a filing cabinet.” Correct this. Explain why.

Q4. A new user needs to open and sort files. “A user interface replaces all CPU instructions.” Correct this. How does an interface help?

I Do · Read new knowledge

Key vocabulary

Design — Define success criteria, inputs, algorithms and expected outputs before writing code.

Implementation — Build a small working feature using appropriate constructs.

Testing — Record test inputs, expected outputs, actual outcomes and pass/fail results.

Refinement — Use test evidence to correct program faults and improve reliability.

Test case — An input and expected outcome used to check correctness.

Bug — A defect in the program.

New learning

Design: Define success criteria, inputs, algorithms and expected outputs before writing code. Consider this in a ticket-sales mini project.

Implementation: Build a small working feature using appropriate constructs. Consider this in a pupil quiz project.

Testing: Record test inputs, expected outputs, actual outcomes and pass/fail results. Consider this in a school lunch calculator.

Refinement: Use test evidence to correct program faults and improve reliability. Consider this in a sports points tracker.

Worked example

A ticket-sales mini project needs design and implementation. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1].

3. Implementation [1]; Build a small working feature using appropriate constructs. [1].

4. Give a specific point and a linked reason for each feature.

We Do · Practise with guidance

Guided 1

In a pupil quiz project, explain the purpose of design. [2 marks]

Worked method: Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1].

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program faults and improve reliability. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 15 is a representative valid value away from the boundaries.

Q2. B — Within its local scope

Q3. Capturing data during transmission, potentially exposing confidential information.

Q4. A user interface lets people use the computer, through GUI controls or typed commands. Other OS tasks manage devices, accounts, memory and files.

I Do / We Do pages · Guided practice

Q1. Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1].

Q2. Implementation [1]; Build a small working feature using appropriate constructs. [1].

Q3. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1]. Refinement [1]; Use test evidence to correct program faults and improve reliability. [1].

You Do pages · Code First

Expected result: Original output: 15. Updated tests: At 0: total 0; at 4: total 20. A negative value should print an error after validation is added. Write the expected and actual output separately.

You Do pages · Independent questions

Q1. Design [1]; Testing [1].

Q2. Implementation [1]; Build a small working feature using appropriate constructs. [1]. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1].

Q3. Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1]. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1]. Refinement [1]; Use test evidence to correct

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Guided 2

A school lunch calculator needs implementation. Explain its role. [2 marks]

Worked method: Implementation [1]; Build a small working feature using appropriate constructs. [1].

Guided 3

A sports points tracker uses testing and refinement. Explain both. [4 marks]

Worked method: Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1]. Refinement [1]; Use test evidence to correct program faults and improve reliability. [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Trace the OCR ERL program, test a new input and improve the code.

```
ticketPrice = 5
quantity = 3
total = ticketPrice * quantity
print(total)
```

Predict the exact initial output.

Test quantity = 0 and quantity = 4.

Add an IF check that rejects a negative quantity.

Question 1

State TWO concepts relevant to this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain how implementation and testing help in a book-return reminder. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

Explain the roles of design, testing and refinement in a classroom club booking. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Planning a program involves identifying success criteria, inputs, algorithms and expected outputs before coding.